Unit 3 Classwork — Equations & Inequalities

One/Two-Step · Distributive · Both Sides · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1 Score 3

Solve: $4x = 28$

Answer: _____

Problem 2 Score 3

Solve: $3x + 5 = 20$

Answer: _____

Problem 3 Score 3

Solve: $\frac{x}{3} - 2 = 4$

Answer: _____

Problem 4 Score 4

Solve: $-2x + 7 = 1$

Answer: _____

B. Distributive Property & Variables on Both Sides

Problem 1 Score 3

Solve: $3(2x - 1) = 15$

Answer: _____

Problem 2 Score 4

Solve: $7x - 2 = 3x + 14$

Answer: _____

C. Multi-Step Equations

Problem 1 Score 4

Solve: $3(x + 2) = 2x + 10$

Answer: _____

Problem 2 Score 4

Solve: $5 - 2x = 3x + 15$

Answer: _____

Problem 3 Score 5

Solve: $4x - 3 = 2(x + 5)$

Answer: _____

Problem 4 Score 5

Solve: $\frac{2}{3}(x + 6) - 2 = x + 4$

Answer: _____

D. Special Cases — No Solution or Infinite Solutions

Problem 1 Score 4

Solve: $2x + 5 = 2x - 1$ — how many solutions?

Answer: _____

Problem 2 Score 5

Solve: 3(x + 4) = 3x + 12 — how many solutions?

Answer: _____

Problem 3 Score 5

Why does 3(x + 2) = 3x + 10 have no solution?

A. Both sides have 3x, so every number works.
B. It does have a solution: x = 4/3.
C. An equation with parentheses never has a solution.
D. After you distribute, both sides have 3x, but 6 does not equal 10.

Answer: _____

E. Linear Inequalities

Problem 1 Score 3

Solve: 3x ≤ 21

Answer: _____

Problem 2 Score 3

Solve: 5x − 3 ≥ 12

Answer: _____

Problem 3 Score 4

Solve: −2x > 8 (careful!)

Answer: _____

Problem 4 Score 5

Solve: −3x + 1 ≤ 10 (careful!)

Answer: _____

F. Word Problems

Problem 1 Score 4

A taxi charges $3 plus $2 per mile. If the total fare was $17, how many miles were driven?

Answer: _____

Problem 2 Score 5

A gym membership costs $25 to join plus $15/month. You can spend at most $100. What's the max number of months?

Answer: _____

Problem 3 Score 5

Plan A: $25 to join a gym plus $15 each month. Plan B: $20 each month with no joining fee. After how many months do the two plans cost the same?

Answer: _____